



Yulong Ge

Ph.D. Student in Applied Mathematics

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Education

Ph.D.	Harbin Institute of Technology , Applied Mathematics • Supervised by Professor Jianwei Ma. • Research focuses on computer vision, generative models, and neural network interpretability.	Harbin, China
B.S.	Harbin Institute of Technology , Information and Computing Science	Harbin, China
Minor	Harbin Institute of Technology , Big Data Management and Applications	Harbin, China

Publications

Feature-Space Planes Searcher: A Universal Domain Adaptation Framework for Interpretability and Computational Efficiency Feature-space unsupervised domain adaptation with frozen encoders, decision-boundary adaptation, and efficient full-dataset optimization. Zhitong Cheng, Yiran Jiang, Yulong Ge, Yufeng Li, Zhongheng Qin, Rongzhi Lin, Jianwei Ma 10.1109/TPAMI.2026.3703974	June 2026
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Projects

Feature-Space Planes Searcher (FPS) for Unsupervised Domain Adaptation

Feature-space unsupervised domain adaptation framework that keeps pretrained encoders frozen and adapts decision boundaries using reusable H5 feature banks.

- Public PyTorch implementation for efficient and interpretable feature-space domain adaptation.
- Related work published in IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Early Access, 2026.

Intelligent Identification and Analysis of Martian Geological Faults

AI for Science project for identifying geological faults in Martian data.

- Developed core components of an AI fault-recognition pipeline.
- Vectorized and standardized fault data for downstream analysis.

Efficient Sampling Algorithms for Diffusion Generative Models

Undergraduate thesis on improving the sampling efficiency of diffusion models.

- Investigated and designed algorithms targeting a central efficiency challenge in diffusion-model sampling.

Generative-Model-Based Ultrasound Medical Image Translation

Cross-modal medical-image translation research using generative models.

- Explored generation of diagnostically informative images to address limitations of single-modality ultrasound data.

Technical Skills

Mathematics and Theory: Computational mathematics, applied mathematics, linear algebra, probability, statistical learning theory, variational inference

Large Language Models: Foundations of LLM fine-tuning and practical LoRA fine-tuning

Generative Modeling: Diffusion models for natural-image generation

Embedded Systems: STM32 development and SolidWorks mechanical modeling

Languages: Chinese (native); English (CET-6 579, CET-4 593)

Selected Honors

- People's Scholarship; Outstanding Youth League Member; Outstanding Student; Special Graduate Academic Scholarship

Volunteer Experience

- **Computer 110 Club (2020-2022):** Supported on-campus computer diagnostics and wrote new-semester purchasing guides for students.